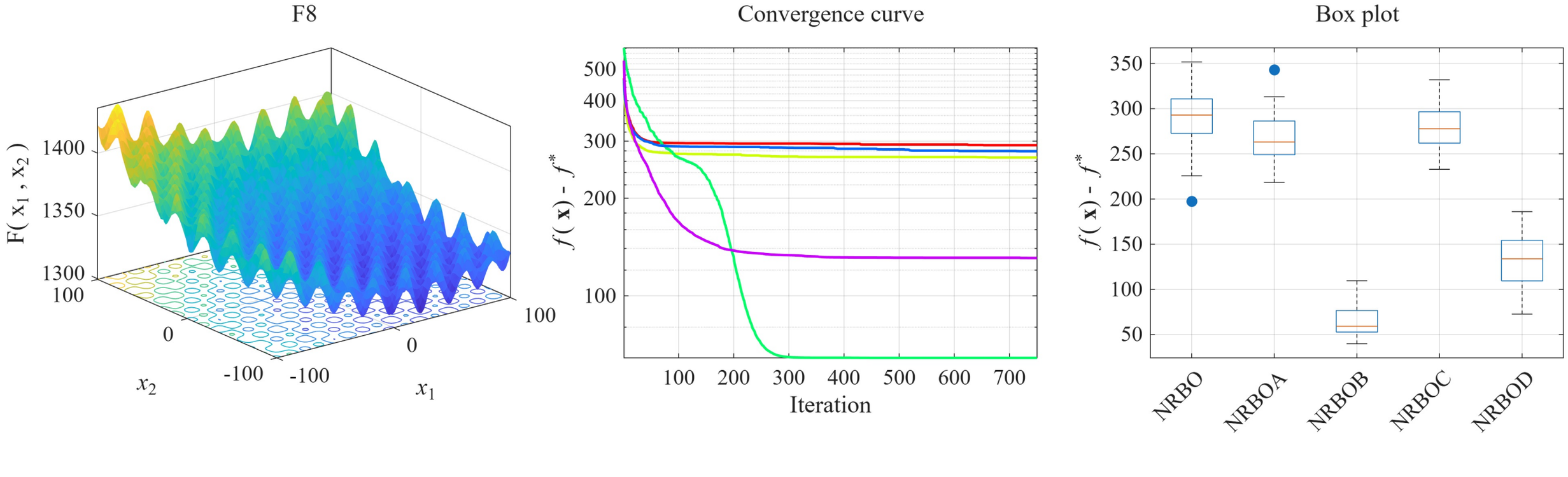

F8
F(x1 , x2)
1400
1350
1300
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
500
400
300
200
100
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
350
300
250
200
150
100
50
NRBO
NRBOA
NRBOB
NRBOC
NRBOD
NRBO
NRBOA
NRBOB
NRBOC
NRBOD